

Clinical Trial Accessibility and Disease Burden Analysis in the United States

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Motivation

Clinical trials are often uneven across the United States, while disease burden is also distributed unevenly. A state can have a large number of residents affected by a condition but still have relatively little trial activity for that condition. This project asks a simple but important question: "Are clinical trials proportionally available in regions and disease categories where they are most needed?" To answer that, I combined three public data sources into one integrated state-disease dataset: ClinicalTrials.gov records for trial availability, CDC Chronic Disease Indicators for burden measures, and state population data for normalization. Comparing trial availability to disease burden at the same geographic level reveals possible mismatches that matter for public health planning, patient access, and research investment. When patients in high-burden regions cannot access trials, they lose the opportunity to receive investigational treatments that may be more effective than standard care, and researchers lose the diverse populations needed to produce generalizable findings. Understanding where these gaps exist is a necessary first step toward making clinical research more equitable and ensuring that trial placement decisions are informed by actual need rather than institutional convenience.

Description of Data Sources

1. ClinicalTrials.gov: Queried via the ClinicalTrials.gov API using condition-based search terms for each of the nine disease categories. The resulting CSV contains study-level fields including NCT ID, title, search_condition, and a JSON-formatted locations field. Records were parsed, exploded by site, and filtered to U.S. locations to produce a state-level view of trial availability by disease group.
2. CDC Chronic Disease Indicators (CDI): Downloaded as a CSV from the CDC data portal. Provides state-level indicators with fields including Topic, Question, DataValue, confidence limits, and year range. Filtered to disease topics overlapping with the trial dataset, with one representative question selected per disease group for consistent cross-state comparison.
3. State population data: Obtained as a CSV from the U.S. Census Bureau. Used to normalize trial counts into trials per million residents so that large states are not artificially ranked higher than smaller ones. Cleaned to retain only the 50 U.S. states with standardized two-letter abbreviations for joining.

The three raw datasets varied significantly in size. The CDI dataset contained 309,216 rows spanning multiple topics, years, stratifications, and question types. The ClinicalTrials.gov file contained 20,442 study-level records, each with a nested locations field that expanded into many more rows once parsed. The population table contained 50 rows after removing aggregate region entries.

The reduction to 347 final rows reflects deliberate layered filtering: trial site rows were aggregated to one count per state-disease pair, CDI was filtered to one representative question per disease group retaining only the most recent observation, and the three tables were inner-joined on state_abbr and disease_group so only fully matched pairs were kept. COPD and Chronic Kidney Disease were dropped when no clean overlap existed across all three sources, leaving seven disease groups. The population table was added via a left join solely for normalization.

Note: URLs for data sources have been listed at the end of the report.

Data preparation notes: COVID-19 was excluded from the condition mapping at the keyword stage and does not appear in any downstream table.

Key dataset facts

Item	Value
Final integrated rows	347
States covered	50
Disease groups in final merge	7

Integrated Data Model

The data model is unchanged from the Phase 2 submission except that COPD and Chronic Kidney Disease were removed after integration confirmed no clean overlap, reducing the disease count from nine to seven groups. The main join keys are state_abbrev for linking all three sources and the combination of state_abbrev and disease_group for linking trial counts to CDI burden records.

The final integrated table contains one row per state-disease pair with columns for trial_count, trials_per_million, burden_value, burden_unit, burden_value_type, question, year_start, year_end, population, and confidence limits. This structure keeps access and burden comparable in one place while preserving source metadata needed for interpretation.

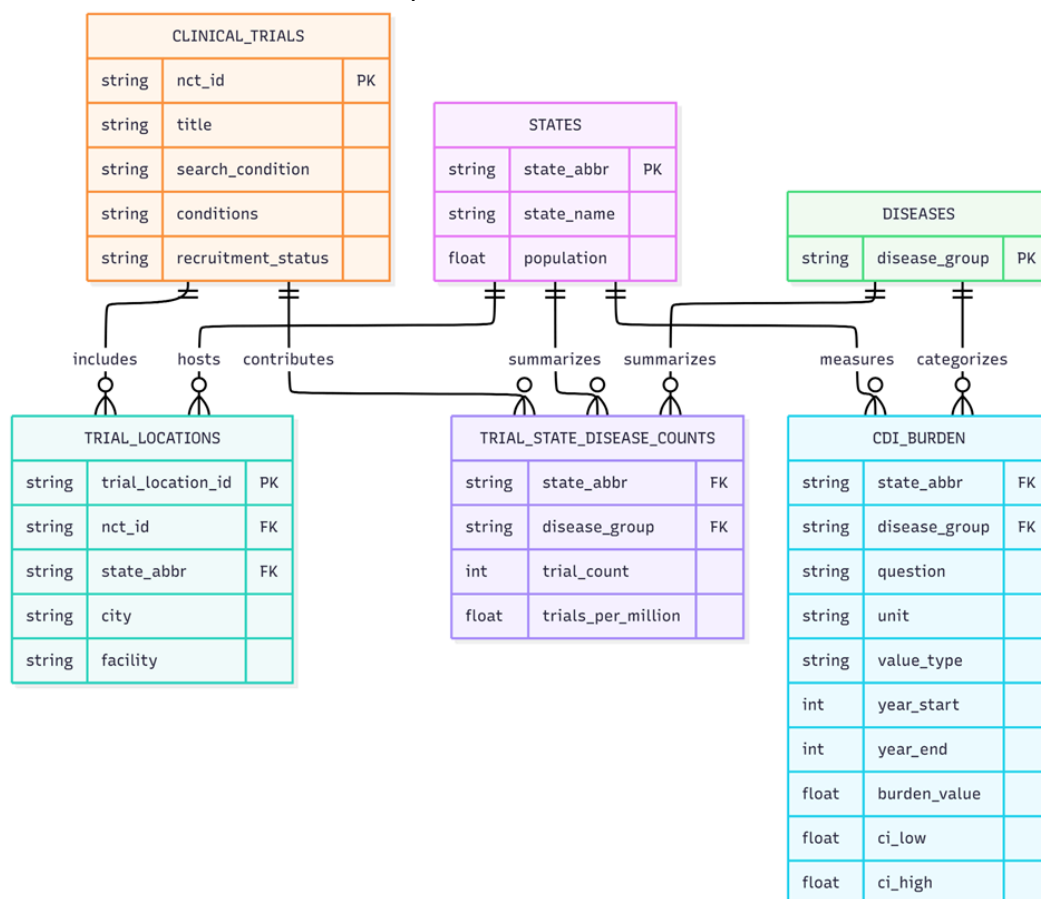


Figure 1. Normalized entity-relationship view for organizing the combined trial, burden, state, and disease data.

Data Cleaning, Analysis, and Visualizations

Analysis Techniques Used

All processing used pandas and numpy, with json for parsing nested location data and pathlib for file handling. The pipeline covered three stages: data cleaning and transformation (parsing JSON locations, filtering to U.S. records, standardizing state abbreviations, mapping free-text conditions to disease categories, and removing duplicates); external data integration (filtering CDI to relevant questions and retaining the most recent state-disease observation, removing aggregate rows from the population table); and derived metrics (trials per capita for normalization, and a z-score based mismatch score to compare burden against trial availability).

The analysis addressed four questions: which disease categories have the most trials, how trial availability varies across states, whether high-burden regions receive proportional trial attention, and which diseases show the largest mismatch between burden and availability.

Figures and Visualizations

All visualizations were created using matplotlib and seaborn.

1. Total Clinical Trials by Disease Group

- Created using a bar chart.
- X-axis: Disease categories
- Y-axis: Number of clinical trials
- Shows that cancer dominates the trial landscape, with other conditions (mental health, cardiovascular disease, diabetes, etc.) having significantly fewer trials.
- This highlights an imbalance in research focus across diseases.

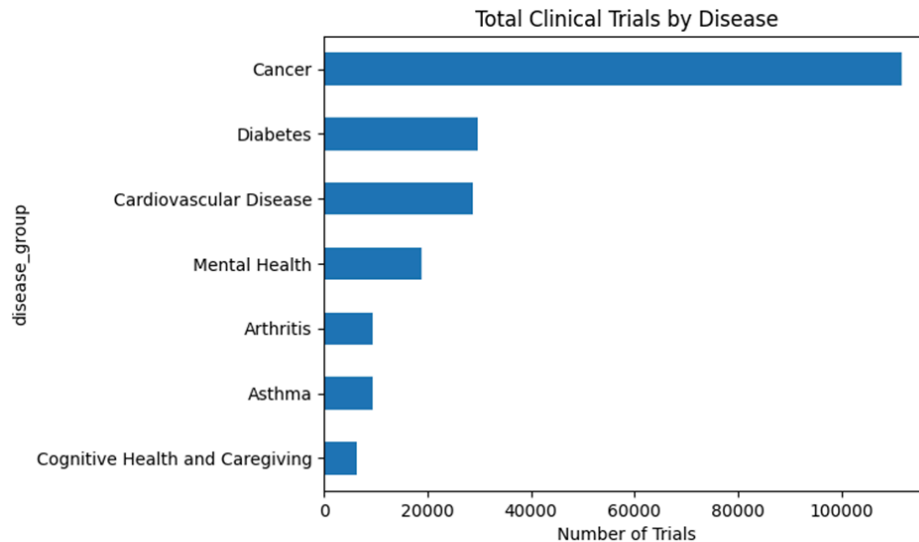


Figure 2. Total clinical trials by disease group. Cancer dominates the trial landscape by a wide margin, while mental health, cardiovascular disease, diabetes, arthritis, asthma, and cognitive health follow at much smaller levels.

2. Clinical Trials per Million People by State

- Created using a bar chart normalized by population.
- X-axis: States
- Y-axis: Trials per million people
- Computed by dividing total trials by state population.
- Reveals that smaller states rank higher per capita, while larger states drop in ranking.
- Demonstrates the importance of normalization when comparing regions.

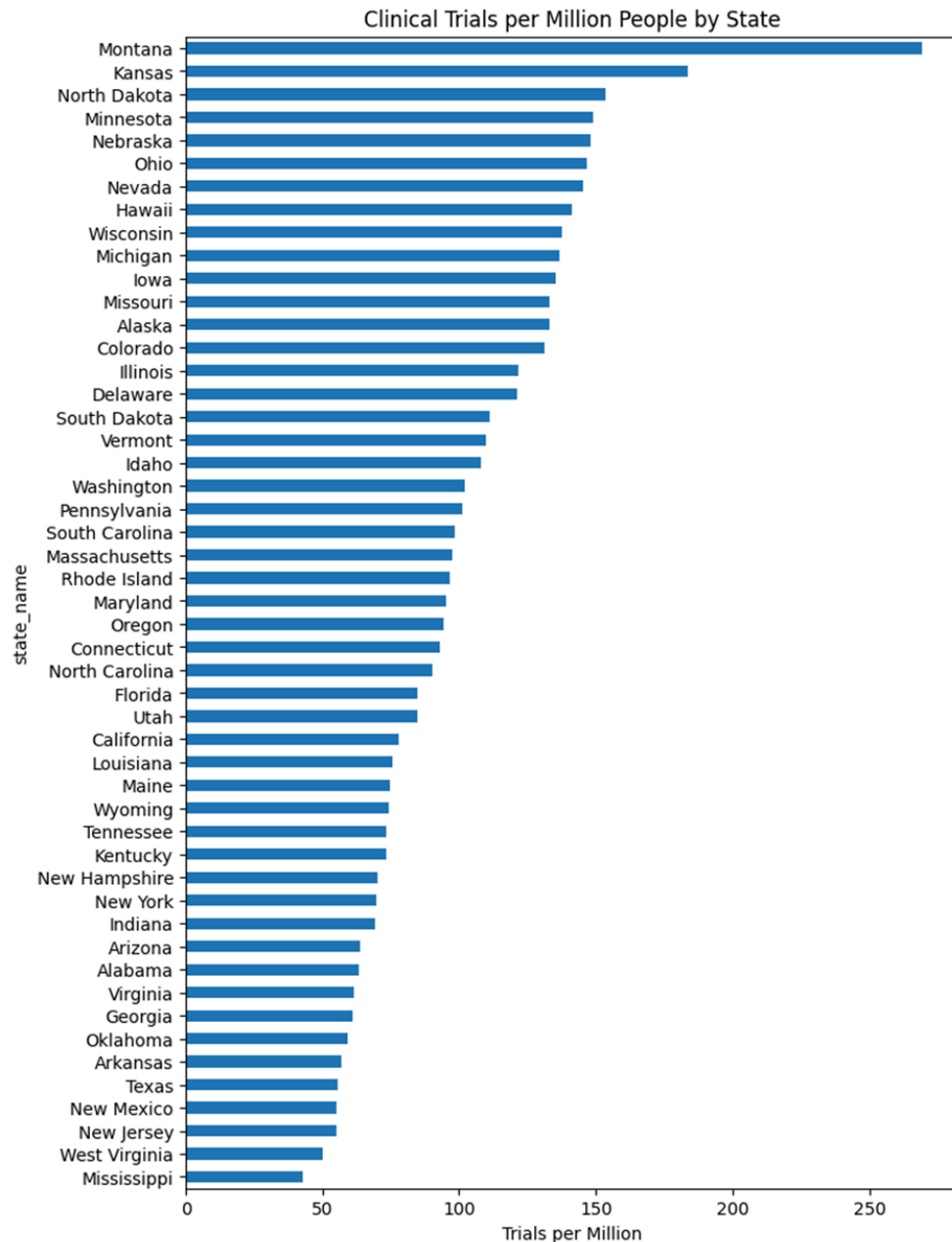


Figure 3. Clinical trials per million people by state. The per-capita view changes the ranking substantially; smaller states such as Montana, Kansas, and North Dakota appear near the top, while several larger states fall lower once population is considered.

3. Disease Burden vs. Trial Availability

- Created using a scatter plot.
- X-axis: Disease prevalence (burden)
- Y-axis: Trials per capita
- Each point represents a state-disease pair.
- Highlights inefficiencies in resource allocation.
- The scatter plot reveals a dense cluster of state-disease pairs concentrated at low burden and low trial density, with a small number of high-burden outliers spread across a wide range of trial values. The distribution does not follow a consistent linear pattern. This is consistent with the z-score normalization used in the mismatch analysis: if burden strongly predicted trial placement, high-burden pairs would consistently appear in the upper right of the chart, but they do not. The spread suggests that factors beyond disease burden, such as the presence of academic medical centers, state population density, or historical research investment, influence where trials are placed.

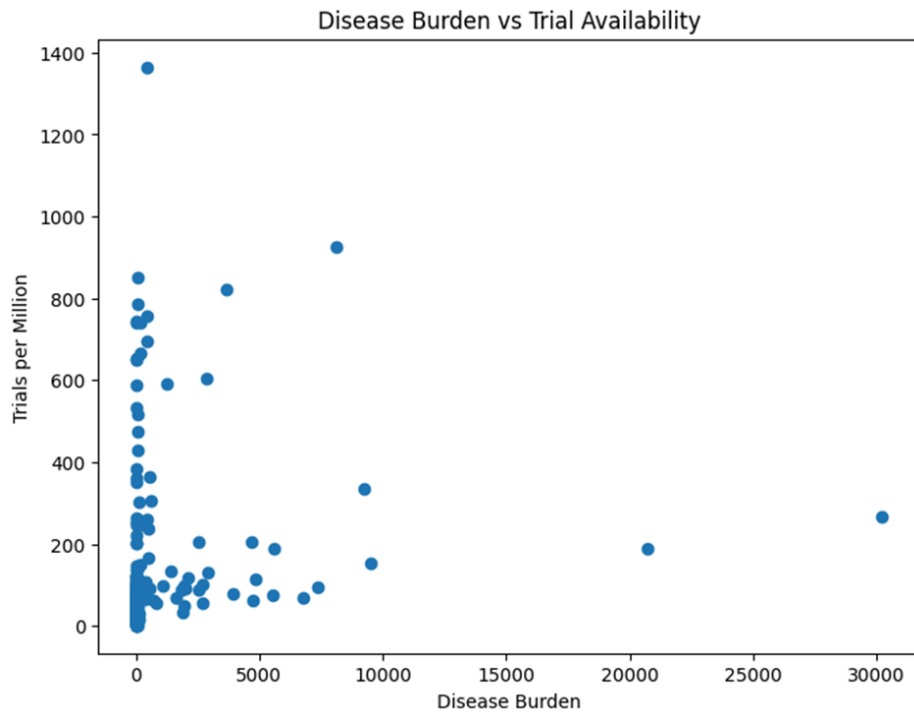


Figure 4. Disease burden versus trial availability. The scatterplot shows a dense cluster near low burden and low trial density, plus a few high-burden outliers. The relationship is not strongly linear, suggesting that burden alone does not determine where trials are placed.

4. Top Mismatches Between Burden and Trial Availability

The mismatch score was computed by first z-score normalizing both the `burden_value` and `trials_per_million` columns across all state-disease pairs (subtracting the mean and dividing by the standard deviation). The mismatch score for each pair is then `burden_norm - trial_norm`. A high positive score means a state carries above-average disease burden for that condition relative to what would be expected, while receiving below-average trial coverage relative to other state-disease pairs. This normalization puts both variables on the same scale so that diseases measured in different units (e.g., cancer incidence per 100,000 vs. prevalence percentages for diabetes) can be compared in a single ranking.

- Created using a horizontal bar chart of mismatch scores, ranked in descending order.
- Displays combinations of state and disease with the largest gaps.
- Example findings:
 - North Carolina - Cancer
 - Virginia -Cancer
 - Arkansas -Cancer
 - Tennessee -Cardiovascular Disease
 - New York -Diabetes
- These represent areas with high disease burden but relatively low trial availability.

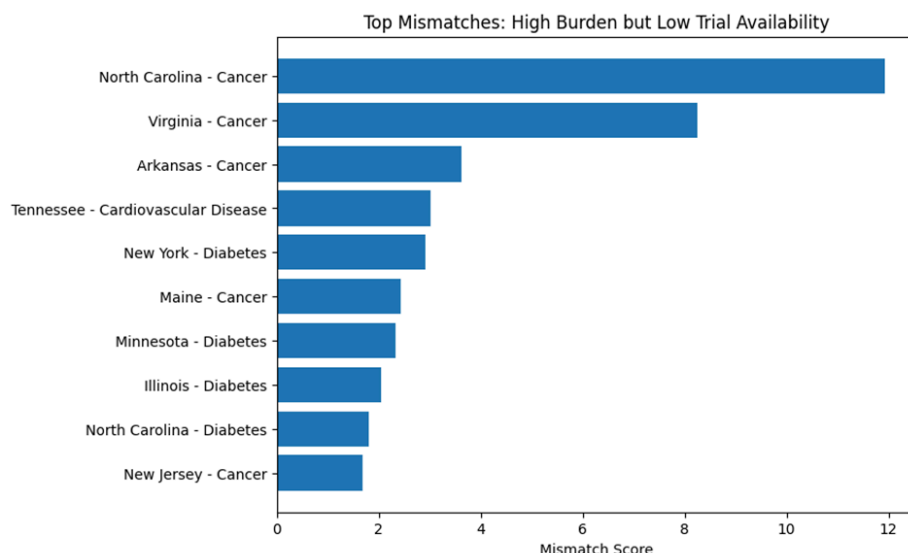


Figure 5. Top mismatches between burden and trial availability. North Carolina-Cancer is the largest mismatch in the final table, followed by Virginia-Cancer, Arkansas-Cancer, Tennessee-Cardiovascular Disease, and New York-Diabetes.

Conclusions

This project analyzes the relationship between clinical trial availability and disease burden across U.S. states by integrating data from ClinicalTrials.gov, CDC CDI, and population datasets. The goal was to understand whether clinical trial distribution aligns with public health needs.

Key findings from the analysis include:

1. Clinical trial activity is highly uneven across diseases, with cancer receiving the largest share of trials in the dataset.
2. There is a significant difference between total trial counts and per-capita availability, showing that population size strongly affects access to trials.
3. Some states with high total trial counts do not rank as highly when adjusted for population, indicating limited relative access.
4. The comparison between disease burden and trial availability reveals clear mismatches, where certain states have high disease burden but relatively low clinical trial presence.

There are also several limitations to consider:

1. Trial counts are based on location data, meaning they reflect trial presence rather than effectiveness or outcomes.
2. CDI indicators vary in units and time periods, which may affect direct comparability across diseases.
3. The disease mapping process relies on free-text matching, which may introduce minor inaccuracies.

Despite these limitations, the integrated dataset provides a meaningful and structured way to analyze geographic disparities in clinical trial access and offers a strong foundation for further research.

Returning to the central question of whether clinical trials are proportionally available where they are most needed, the evidence suggests they are not. Trial placement appears to follow existing research infrastructure more than disease burden patterns. States with high burden for cancer, cardiovascular disease, and diabetes are not consistently among those with the most trial activity per capita. The integrated dataset built here provides a reusable, structured foundation for future work to investigate the causes of these gaps and to track whether the distribution improves over time.

Future Work

Given more time, I would extend the pipeline in four directions: First, I would automate data acquisition directly from the public APIs so the project refreshes from source instead of relying on saved CSVs. Second, I would add more diseases and multi-year trend analysis so that changes over time can be studied instead of only a single snapshot. Third, I would add correlation or regression analysis, possibly with confidence intervals, to quantify whether trial availability is associated with burden after adjusting for population size. Finally, I would create an interactive map or dashboard so the state patterns can be explored visually. The current modular script structure already supports these changes because the cleaning, integration, and analysis stages are separated into distinct steps.

Data Source References

1. ClinicalTrials.gov API: <https://clinicaltrials.gov/data-api/api>
2. ClinicalTrials.gov access overview: <https://clinicaltrials.gov/data-api/how-access-data>
3. CDC Chronic Disease Indicators: <https://data.cdc.gov/Chronic-Disease-Indicators/U-S-Chronic-Disease-Indicators/hksd-2xuw>
4. CDC data portal: <https://data.cdc.gov/>
5. U.S. Census Bureau population estimates and developer resources: <https://www.census.gov/data/developers/data-sets/popest-popproj/popest.html>